TAAPSHAMAN: Thermal Resilience Scoring and Retrofit Measures for Buildings

Aditya Singh Chauhan
26M2067
Centre for Climate Studies, IIT Bombay

Utkarsh Yadav
26M2066
Centre for Climate Studies, IIT Bombay

Archishman Roy
26M2068
Centre for Climate Studies, IIT Bombay

Aarya Kokate
23B3901
Dept. of Electrical Engineering, IIT Bombay

Adeep Kuncalienker
23B2261
Dept. of Mechanical Engineering, IIT Bombay

***Abstract*—Heat exposure in Indian buildings is rising, yet most assessments rely on blueprints or arbitrary score weights. TAAPSHAMAN proposes a Thermal Resilience Score (TRS) for existing buildings that links climate exposure and building characteristics to measured indoor thermal response, identifies the main heat-risk sources and compares practical retrofits. We summarise five reviewed studies and present a machine-learning prototype trained on synthetic data. The prototype is a pipeline test only; its metrics do not describe real-world accuracy.**

I. INTRODUCTION

Overheating in existing buildings is driven by solar heat gain, envelope properties, ventilation and the cooling available to occupants. Retrofits such as insulation, shading and solar-gain reduction are well documented, but there is no simple, evidence-based way to rank buildings by thermal resilience or to choose among retrofits without a full simulation. TAAPSHAMAN targets this gap with an existing-building assessment rather than a blueprint-based one, producing a TRS, a list of dominant heat-risk sources and a ranked set of retrofit options. This report covers the initial stage: literature review, framework definition and a prototype scoring pipeline.

II. METHODS

A. Literature review

We reviewed five studies on thermal comfort, building retrofit, thermal adaptation and machine-learning-based overheating prediction in Indian and international buildings. The aim was to learn how thermal performance is measured, which building and climate variables matter, and how retrofit interventions are evaluated. Two studies anchor our approach: an IIT Bombay CUSE building retrofit study [1], which combines measured indoor temperature, model validation and quantitative comparison of retrofit options, and the Indian thermal-comfort analysis of Kumar [2] covering 17,569 observations.

B. Assessment framework

Fig. 1 shows the framework. Climate exposure and building characteristics determine the indoor thermal response; this response reveals where heat risk originates; retrofit scenarios are then compared on the same response; and the TRS summarises the result. The case study is the Rahul Bajaj Building at IIT Bombay.

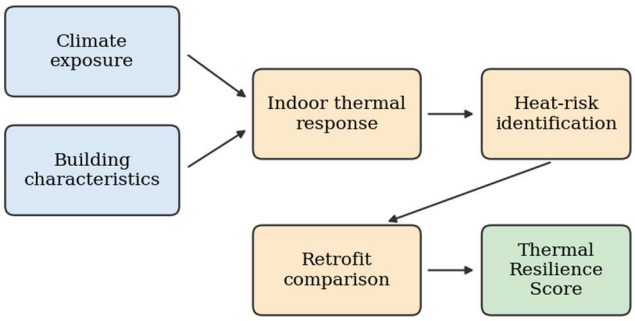


Fig. 1. TAAPSHAMAN assessment framework. Blue: inputs; orange: analysis stages; green: output.

C. Prototype scoring pipeline

As a software test, we generated a 1,000-building dataset with 29 columns. The envelope, weather and indoor-temperature fields were generated randomly from a transparent illustrative relationship; they are not observations. The target was daily maximum indoor temperature. Two other generated indoor outcomes, the building ID and an undocumented satellite feature were excluded to avoid leakage, leaving 13 numeric and 11 categorical predictors. After an 80/20 split (seed 42), median and most-frequent imputation and one-hot encoding were fitted on training data inside the pipeline. Random Forest and Extra Trees regressors (350 trees, minimum leaf size 2) were compared by MAE, RMSE and R^2.

The prototype score maps predicted maximum indoor temperature T linearly between illustrative bounds of 26 and 40 °C: $TRS = 100(40 - T)/14$, clipped to 0–100. The complementary risk index (100 − TRS) is binned as low (<25), moderate (<50), high (<75) or very high. Transparent rules then flag retrofit checks, for example a dark or concrete roof, roof insulation below 20 mm, no external shading, poor ventilation, no air conditioning or tree cover below 15%.

III. RESULTS

A. Evidence from the literature

The CUSE study evaluated 15 envelope-retrofit scenarios; the combined case gave 25% annual energy savings and a 60% reduction in hours above 35 °C [1]. Across the reviewed work, insulation, shading and reduced solar heat gain improved thermal conditions, while data-driven models gave faster building-level predictions than simulation. Kumar's Indian dataset spans air-conditioned, naturally ventilated and mixed-mode buildings and shows that thermal response varies with

ventilation mode, season and building type [2]. Outside India, a roof retrofit of a London office lowered indoor air temperature from 29.3 to 26.9 °C [3], and building characteristics such as insulation strongly influence the propensity to high indoor summer temperatures [4]. Together these support a measured, evidence-based score instead of weights assigned by judgement.

B. Prototype (synthetic data)

Extra Trees performed best (Table I) and was used for scoring. Fig. 2 shows test-set predictions; the model captures the trend but underpredicts the hottest cases. Models for the other two generated outcomes reached R^2 of 0.566 and 0.483.

TABLE I. Held-out test performance (synthetic target)

Model	MAE (°C)	RMSE (°C)	R^2
Extra Trees	0.961	1.231	0.630
Random Forest	1.015	1.302	0.586

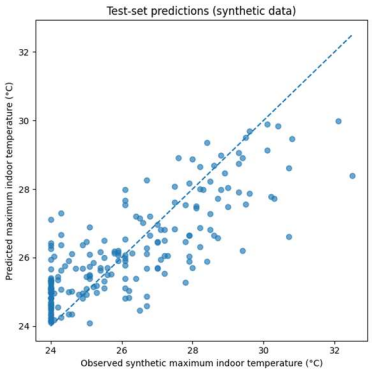


Fig. 2. Predicted vs. synthetic maximum indoor temperature on the test set (Extra Trees). Dashed line: 1:1.

Permutation importance (Fig. 3) ranks outdoor temperature first, followed by air conditioning, window area ratio and roof insulation. Across all 1,000 buildings the highest predicted maximum was 31.9 °C (TRS 57.6, moderate); the fifteen riskiest buildings scored 57.6–67.2, and none reached the high category. The first retrofit flag for 13 of these 15 was roof solar heat gain.

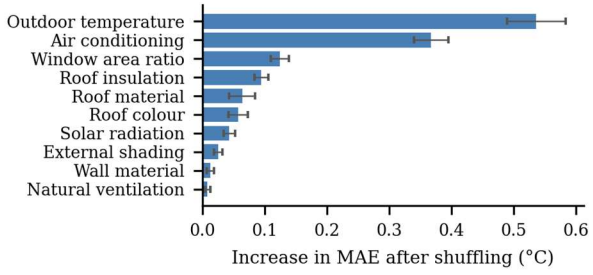


Fig. 3. Permutation importance of the ten most influential features (mean ± s.d., 8 repeats; synthetic data).

IV. DISCUSSION

The prototype shows that the full chain from inputs to a score and retrofit prompts works end to end. It does not show that the model predicts real buildings. Because the target was generated from assumed relationships, the importance ranking (outdoor temperature and air conditioning first) mostly reflects those assumptions and is not a finding about real buildings. Two design weaknesses also appear. Many generated maxima sit at a 24 °C floor and the 26 °C lower bound gives every cooler building a TRS of 100, so the score cannot separate good performers. The 40 °C upper bound is also never approached, leaving the upper categories unused. Thresholds should therefore come from local thermal-comfort guidance, such as the India Model for Adaptive Comfort [5], and from measured data, and hours above a comfort threshold should enter the score, as reported in [1]. Related probabilistic heat-risk and machine-learning studies [6], [7] point the same way.

V. NEXT STEPS

We will (i) finalise measurable parameters, (ii) build an evidence-based scoring method, and (iii) tie the score to retrofit recommendations for the Rahul Bajaj Building. We will log indoor and outdoor conditions there, relate them to local weather observations, replace every synthetic column with observed data, and recalibrate the score bounds against these measurements.

SUPPLEMENTARY INFORMATION

The analysis notebook (TAAPSHAMAN_Final.ipynb), figures and result tables accompany this report. The synthetic data are illustrative only.

REFERENCES

[1] IIT Bombay CUSE building envelope-retrofit study: 15 scenarios, measured indoor temperature and model validation.

[2] Kumar (2022), analysis of 17,569 thermal-comfort observations in Indian AC, naturally ventilated and mixed-mode buildings.

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[6] Bianchi and Ciurlanti (2026), probabilistic PBEE-based heat-risk framework for buildings.

[7] Akyol et al. (2025), machine-learning UBEM prediction of heating energy and overheating under climate change.